Ivy Li

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science, Honours, Co-operative Program — *Sep. 2024 – Apr. 2029*

- **GPA**: 4.0/4.0, 95.2/100
- **NSERC Undergraduate Student Research Award**: Canadian federal award supporting student research.
- **Rene Descartes National Mathematics Scholarship ($25,000)**: Awarded to 15 Mathematics faculty students annually.

Skills

Core Languages: Python, C++, C, C# TypeScript, JavaScript, Java, Scala, HTML, CSS, SQL
Frameworks: Django, React, Node.js, NextJS, FastAPI, Express, LangChain, PyTorch, Tensorflow, Scikit-learn, Pandas
Tools: Git, Unix/Shell Scripting, Google Cloud Platform, Docker, PostgreSQL, MongoDB, Selenium, Jupyter, Unity
Certifications: NVIDIA Generative AI LLMs

Experience

Remote Sensing AI Engineer Intern, A.U.G. Signals — May 2026 – Aug. 2026

- Developing **ML pipelines** in Python to process remote sensing data and predict harmful algae bloom growth
- Engineering a full-stack **web GIS platform** with asynchronous task processing using Django, Celery, Redis, and Docker

Human-Computer Interaction Research Assistant, UWaterloo WVisdom HCI Lab — May 2025 – Aug. 2025
LLM-Driven Camera Systems in VR — *Supervised by Prof. Jian Zhao*

- Optimized multi-agent workflows in Python to **improve system throughput by 4x**
- Designed and deployed a **model context protocol (MCP) server** and **function tooling** for an AI camera agent in VR, reducing hallucination rates and improving agent reliability and correctness
- Engineered a **full-stack web system** under the MVC framework with a **voice-controlled user interface** and **multimodal input support** using React, FastAPI, and Pydantic-ai, integrating with OpenAI API for AI-powered VR streaming
- Developed an **end-to-end testing framework** for WebVR using Selenium to support regression testing and rapid iteration

Eye-tracking for Pilot Training — *Supervised by Prof. Jian Zhao and Prof. Ewa Niechwiej-Szwedo*

- Devised an automatic eye-tracking calibration workflow using unsupervised learning, **reducing calibration time by 90%**
- Built **scalable data pipelines** in Python (Pandas, Matplotlib, Scikit-learn) to analyze large eye-tracking datasets
- Developed an **interactive GUI** for eye-tracking data visualization and analysis in Dash

Software Developer, Waterloo Reality Labs — Nov. 2024 – Apr. 2025

- Built and iterated on a **VR captioning system**, shipping end-to-end features to generate rich text descriptions of 3D scenes
- Implemented **hand gesture input systems** in Unity (C#) and integrated them into a larger application architecture

Web Developer Assistant, U+ Education — Feb. 2024 – Jun. 2024

- Led a team of **4 interns** to build, ship, and maintain a **production web crawler** to aggregate data from **1100+ student-led organizations** across **10+ universities**

Projects

VM Text Editor | *C++, Ncurses* — Dec. 2025

- Built a full function **text editor** modelled after Vim, supporting **50+ reversible commands** and macro creation
- Implemented a **gap-buffer data structure** to maintain responsive editing performance as document size grows

Goose on the Loose | *TypeScript, NextJS, Mongo, Vercel, Git* — Sep. 2024

- Built and shipped a **location-based interactive game** inspired by Pokemon Go, collaborating in a team of 4 to deliver a **deployable web application** using NextJS

NewSight: A News Bias Detector | *Python, JavaScript, React, Node.js, Express, Google Cloud, Docker, Git* — Aug. 2024

- Developed a Chrome extension with a **client-server architecture**, integrating **third-party APIs** to detect bias in the media
- Built a **Dockerized back-end** in ExpressJS with Python services for scalable data processing and REST API integration

Awards

Major League Hacking Award: Hack the North 2024 sponsor prize awarded to 1 out of 40 projects. — 2024
2x American Mathematics Contest (AMC 12) Honour Roll: Top 5% of AMC 12 participants internationally — 2023
Euclid Mathematic Contest Honour Roll: Top 5% of Euclid participants internationally — 2023